

Widest-Path vs. Shortest-Path Routing Under CSMA/Bianchi WiFi Interference

ncsim Experiment Report

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Abstract

We compare **widest-path** and **shortest-path** multi-hop routing strategies for DAG-scheduled task graphs on wireless mesh networks using the ncsim discrete-event simulator. Under the CSMA/Bianchi interference model with HEFT scheduling, we evaluate 9 combinations of network size (2×2 , 3×3 , 4×4 grids) and DAG complexity (5, 10, 20 tasks). Our results show that widest-path routing consistently underperforms shortest-path routing in 6 of 9 cases, with slowdowns exceeding $35\times$ on larger networks. The root cause is *flow concentration*: widest-path funnels multiple data transfers onto the same high-bandwidth links, creating severe contention under the CSMA/Bianchi MAC model, while shortest-path distributes traffic across shorter, more diverse routes.

1 Introduction

When scheduling DAG-structured computations across a wireless mesh network, data must be transferred between nodes after each task completes. The *routing strategy* determines which multi-hop path each transfer takes. Two natural choices are:

- **Widest path**: maximize the minimum link bandwidth along the route (Dijkstra-style, maximizing bottleneck capacity).
- **Shortest path**: minimize the number of hops (BFS-style, minimizing hop count).

In isolation, widest-path routing selects higher-throughput links. However, in a wireless network where all links within carrier-sensing range contend for airtime via CSMA, concentrating flows on fewer “best” links can create severe contention. This report quantifies this effect using ncsim’s CSMA/Bianchi interference model.

2 Experimental Setup

2.1 Network Topologies

We use square grid mesh networks with 40 m spacing between adjacent nodes. Links include all grid edges (horizontal, vertical) and checkerboard-pattern diagonals, all bidirectional. The WiFi PHY rates are computed from the 802.11ax MCS table based on SNR at each link distance: 40 m grid links and ≈ 56.6 m diagonal links receive different PHY rates, giving widest-path and shortest-path different preferred routes.

Nodes have heterogeneous compute capacities (80–300 compute_units/s) so that HEFT distributes tasks across multiple nodes, creating inter-node data transfers.

Table 1: Network configurations.

Label	Grid	Nodes	Bidirectional Links
Small	2×2	4	12
Medium	3×3	9	32
Large	4×4	16	68

2.2 DAG Structures

Three DAG sizes capture different communication patterns:

Table 2: DAG configurations (all tasks: compute_cost=500, all edges: data_size=10 MB).

Label	Tasks	Shape
Small	5	Fork-join: 1 source $\rightarrow$ 3 parallel $\rightarrow$ 1 sink
Medium	10	Diamond: source $\rightarrow$ 4 $\rightarrow$ 4 $\rightarrow$ sink
Large	20	Multi-level: 5-stage pipeline with branching

2.3 Simulation Parameters

All simulations use:

- **Scheduler:** HEFT (Heterogeneous Earliest Finish Time)
- **Interference:** CSMA/Bianchi (802.11ax, 5 GHz, 20 dBm TX, path loss exponent 3.0)
- **Carrier sensing range:** 71.2 m (all links within each grid are in the conflict graph)
- **Seed:** 42 (deterministic)

The CSMA/Bianchi model combines two interference mechanisms: (1) *Bianchi contention* for links sharing a conflict-graph edge (symmetric MAC-layer time-sharing), and (2) *SINR degradation* for hidden terminals outside the conflict graph. In our grid topologies, nearly all links are within carrier-sensing range, so Bianchi contention dominates.

3 Aggregate Results

The dominant trend is clear: as network size increases, widest-path routing degrades dramatically. On the 4×4 grid, widest-path is $35\times$ slower for the small DAG and $5\times$ slower for the large DAG. The single case where widest-path wins (2×2 , medium DAG) occurs because the tiny network offers no path diversity, and widest-path’s link selection happens to balance load better.

4 Per-Scenario Analysis

Each figure below shows the widest-path (left) and shortest-path (right) routing side by side. The top row displays the network topology with links colored and thickened by flow count (number of data transfers using that link). The middle row shows a Gantt-style timeline of task executions (blue) and data transfers (orange). The bottom row shows per-directed-link flow counts.

Table 3: Makespan (seconds) for all 18 simulations. Bold indicates the winner in each cell. Percentage shows widest-path overhead relative to shortest-path.

DAG	Routing	Network Size		
		2×2 (4)	3×3 (9)	4×4 (16)
Small (5)	Widest	11.43	12.37	315.20
	Shortest	11.43	12.37	8.93
	<i>Diff</i>	<i>tie</i>	<i>tie</i>	+3,428%
Medium (10)	Widest	20.07	29.52	1,396.86
	Shortest	26.03	25.00	277.73
	<i>Diff</i>	-23%	+18%	+403%
Large (20)	Widest	47.95	89.44	1,518.28
	Shortest	35.61	62.79	283.20
	<i>Diff</i>	+35%	+42%	+436%

Table 4: Winner summary: shortest-path wins in 6 of 9 cases.

	2×2	3×3	4×4
Small DAG	Tie	Tie	Shortest
Medium DAG	Widest	Shortest	Shortest
Large DAG	Shortest	Shortest	Shortest

4.1 2×2 Network $\times$ Small DAG (5 tasks)

With only 4 nodes and a simple fork-join DAG, HEFT places 4 of 5 tasks on the fastest node (n0, capacity=200) and sends one task to n2. Both strategies use the same single-hop direct link (n0 $\leftrightarrow$ n2) for both transfers. There is no routing decision to make: the network is too small for path diversity.

4.2 2×2 Network $\times$ Medium DAG (10 tasks)

This is the *only case where widest-path wins*. With 10 tasks across 3 nodes, widest path distributes 8 transfers evenly: 2 flows each on the four directed links (n0 $\rightarrow$ n1, n0 $\rightarrow$ n2, n2 $\rightarrow$ n0, n1 $\rightarrow$ n0). Shortest path concentrates 3 flows onto n0 $\rightarrow$ n2 while underusing other links. In a tiny network where all links have the same hop count, widest-path’s bandwidth preference accidentally acts as a load balancer.

4.3 2×2 Network $\times$ Large DAG (20 tasks)

With 20 tasks on 3 nodes, the flow-concentration problem emerges. The topology panel shows widest path piling **14 bidirectional flows** onto the n0 $\leftrightarrow$ n2 link (thick dark red), while the n0 $\leftrightarrow$ n1 link carries only 7 (thinner orange). Widest path also introduces 2-hop routes, doubling link occupancy per transfer. Shortest path distributes traffic more evenly across 3 physical links (max 8 flows), with all transfers using direct 1-hop routes. The Gantt chart shows widest path’s transfers stretching to ~ 45 s while shortest path finishes by 35.6s.

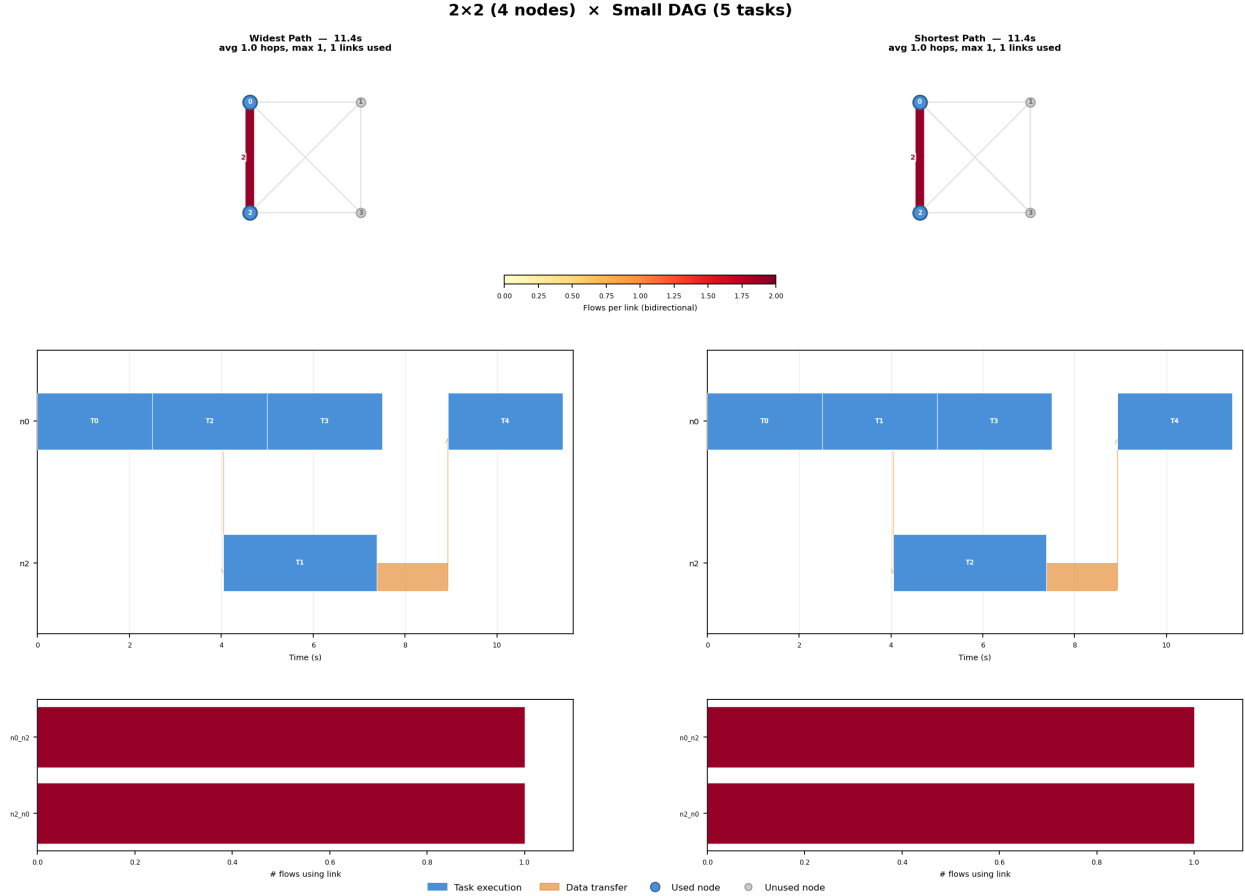


Figure 1: $2 \times 2 \times$ Small DAG. Both routing strategies produce identical results.

4.4 3×3 Network × Small DAG (5 tasks)

HEFT concentrates 4 of 5 tasks on n4 (center, capacity=300) and places one task on a corner node. Both strategies must traverse the same 2-hop path ($n4 \rightarrow n3 \rightarrow n6$ and back). With only 2 sequential transfers and no alternative paths, routing is irrelevant.

4.5 3×3 Network × Medium DAG (10 tasks)

The topology panels diverge clearly. Widest path routes traffic through distant nodes (n0 and n6) via 2-hop paths through n1 and n3, activating **10 unique links**. Shortest path concentrates on n4's direct neighbors (n7 and n6) using only **6 unique links** with a lower average hop count (1.5 vs. 2.0). Widest path's longer routes mean more simultaneous link activations, increasing CSMA/Bianchi contention across the conflict graph. Shortest path finishes 18% faster.

4.6 3×3 Network × Large DAG (20 tasks)

With 20 tasks distributed across 4 nodes, widest path creates a heavily loaded corridor: **6 flows each** on $n4 \rightarrow n3$ and $n3 \rightarrow n6$, plus 4 flows on $n0 \rightarrow n1$ and $n1 \rightarrow n4$. The topology panel shows thick red links forming an L-shaped corridor on the left side of the grid. Widest path also uses 3-hop routes (avg 2.0 hops, max 3), tripling the link occupancy per transfer. The Gantt chart shows n4

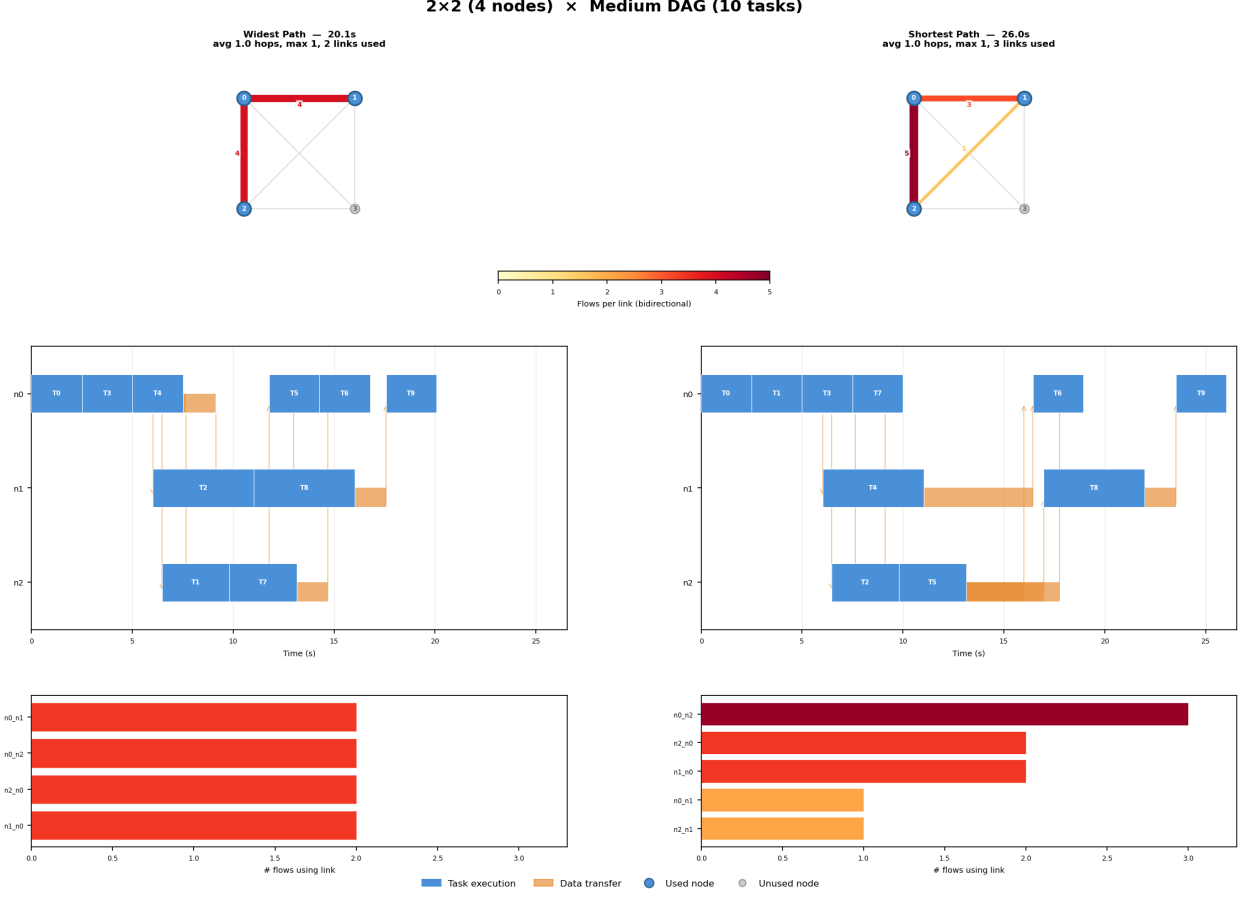


Figure 2: $2 \times 2 \times$ Medium DAG. Widest path wins by balancing load across two links.

blocked by transfer activity from $t = 5$ to $t = 78$ s. Shortest path uses a similar number of unique links (6 vs. 7) but with lower hop counts (avg 1.6, max 2), completing 42% faster.

4.7 4×4 Network x Small DAG (5 tasks)

This is the most dramatic case. HEFT assigns one task to a remote node via widest-path routing: the transfer traverses an **8-hop perimeter route** ($n4 \rightarrow n0 \rightarrow n1 \rightarrow n2 \rightarrow n6 \rightarrow n10 \rightarrow n14 \rightarrow n13 \rightarrow n12$), taking 155s per direction. The topology panel shows all 8 links uniformly red. Each hop adds latency, and with `csma_bianchi`, each of the 8 links must contend for airtime. In contrast, shortest path sends the task to the adjacent node $n0$ via a **single 1-hop link**, completing in 1.5s. The makespan ratio is $315.2/8.9 = 35.4\times$.

4.8 4×4 Network x Medium DAG (10 tasks)

Widest path routes all flows through the same perimeter corridor ($n4 \rightarrow n0 \rightarrow n1 \rightarrow n2 \rightarrow n6 \rightarrow n10 \rightarrow n14 \rightarrow n13 \rightarrow n12$), loading each link with 3–5 concurrent flows. The topology panel shows the entire left and top perimeter glowing uniformly red. With 5 flows sharing 8-hop routes, each individual link sees heavy Bianchi contention, and transfers take ~ 772 s each. Shortest path uses two independent corridors ($n4 \rightarrow n5 \rightarrow n10 \rightarrow n14$ and $n4 \rightarrow n0$), visible as two separate orange paths in the topology panel. Average hops drop from 6.4 to 4.0, and the makespan improves by $5\times$.

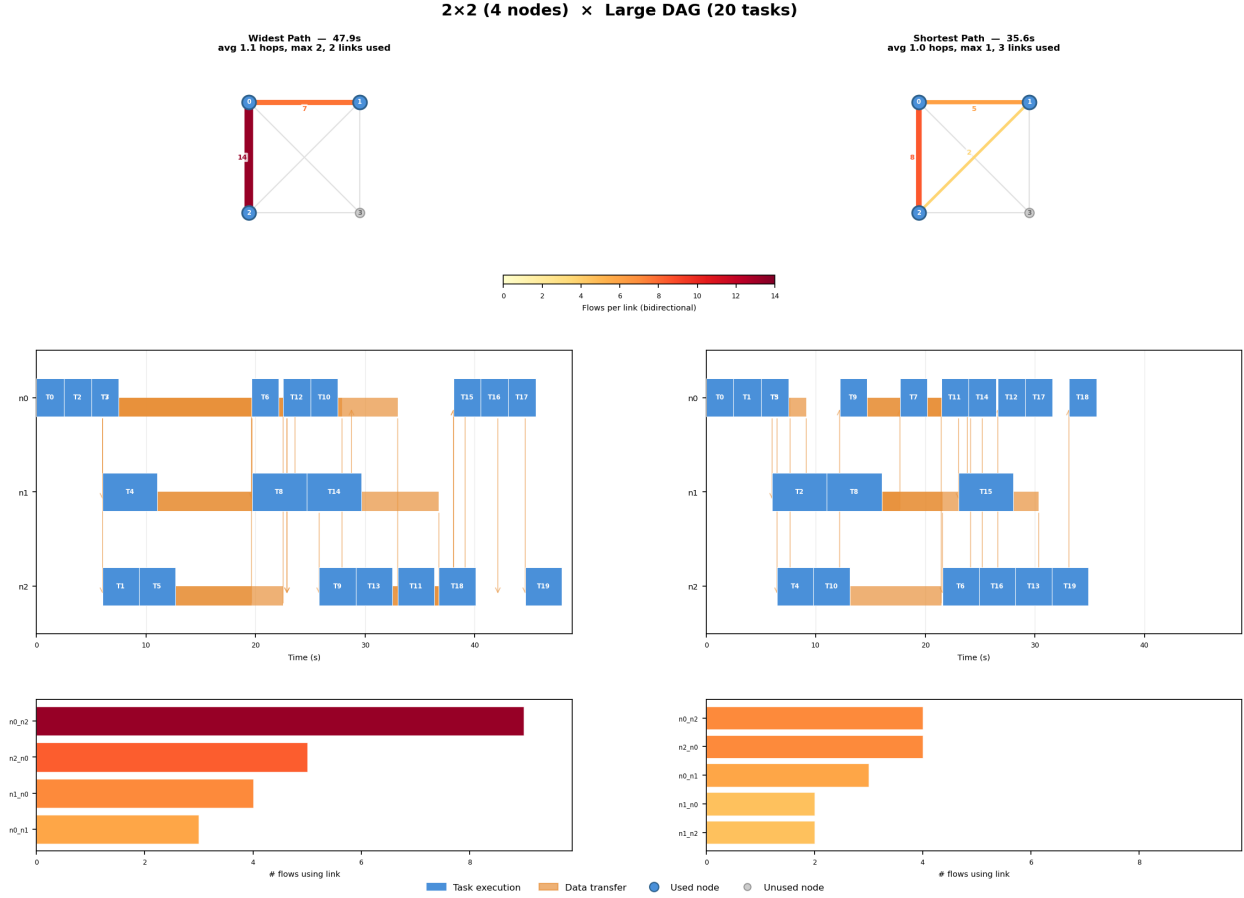


Figure 3: $2 \times 2 \times$ Large DAG. Widest path concentrates 14 flows on link $n0-n2$.

4.9 4×4 Network x Large DAG (20 tasks)

The worst case overall. Widest path routes 18 transfers through the perimeter, loading $n4 \rightarrow n0$ / $n0 \rightarrow n1$ / $n1 \rightarrow n2$ / $n2 \rightarrow n6$ with **7 flows each** (up to 12 bidirectional). The topology panel shows the entire left and top edge as thick dark-red lines, while the right half of the network is completely unused (gray). Average hop count is 5.0, max 8. The first ~ 925 s is consumed entirely by outbound transfers. Shortest path splits traffic across two paths on the right side of the grid ($n4 \rightarrow n5 \rightarrow n10 \rightarrow n14$ and $n0 \rightarrow n1 \rightarrow n2 \rightarrow n6$), with max 5 flows per link and avg 2.7 hops. Makespan: 1,518s vs. 283s ($5.4\times$ improvement).

5 Discussion

5.1 Why Widest Path Fails Under CSMA

Widest-path routing optimizes a *single-flow* metric: it selects the path with maximum bottleneck bandwidth. This is optimal when a single transfer uses the network in isolation. However, DAG execution produces *multiple concurrent transfers*, and widest-path routing directs them all toward the same high-bandwidth links. Under CSMA/Bianchi:

1. **Flow concentration:** Multiple transfers share the same “best” links, dividing effective

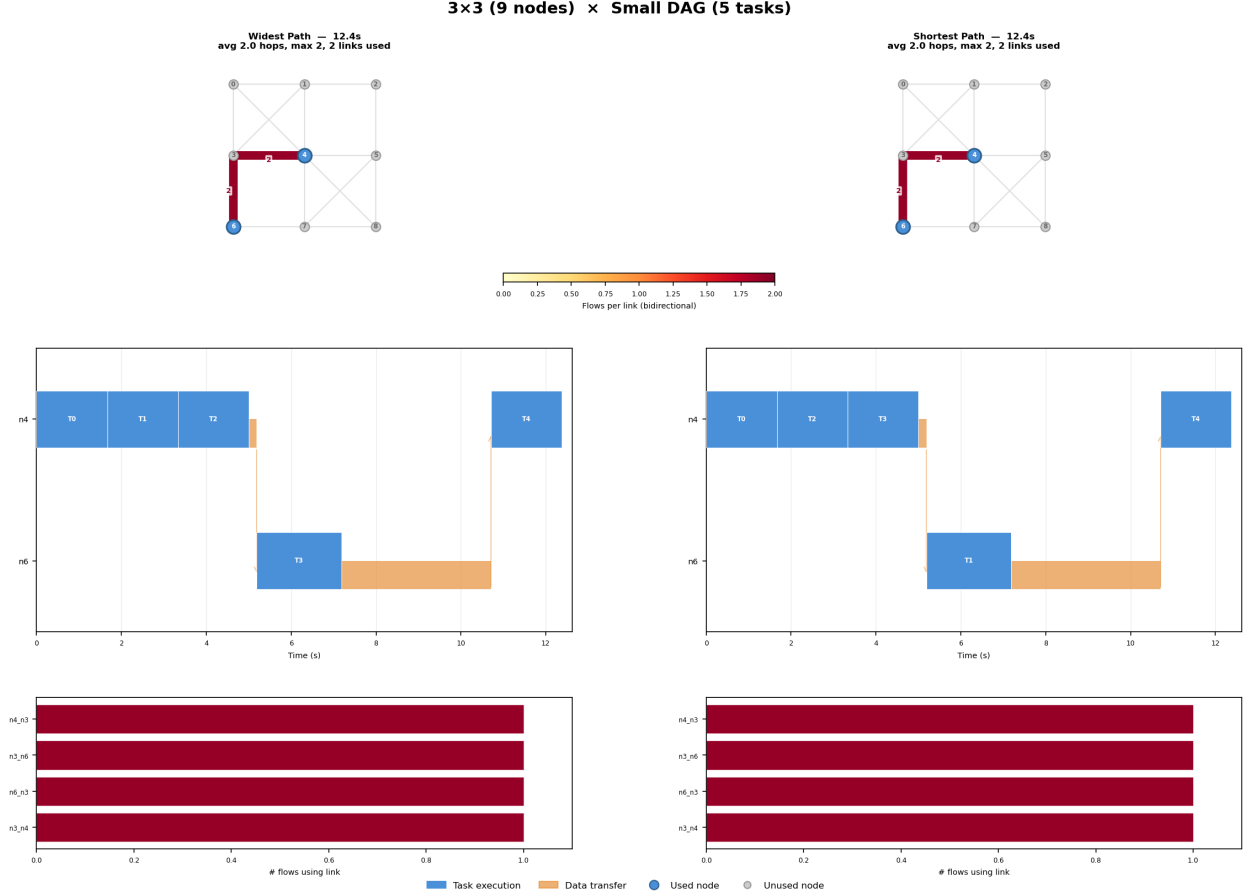


Figure 4: $3 \times 3 \times$ Small DAG. Both strategies take the same 2-hop path through n_3 .

bandwidth by the number of concurrent flows via Bianchi MAC contention ($\eta(n)/n$ scales sublinearly).

2. **Hop count inflation:** To reach distant high-bandwidth links, widest path takes longer routes. Each additional hop means one more link occupied simultaneously, increasing the total contention footprint.
3. **Cascading delays:** Longer transfers block downstream tasks, pushing the critical path further out and amplifying the makespan.

5.2 When Widest Path Wins

Widest path wins only in the $2 \times 2 \times$ Medium DAG case, where:

- The network is so small that all paths have the same hop count (1 hop).
- Widest path’s link preference happens to balance load across the two available physical links better than shortest path.

This is a degenerate case where the “widest” and “shortest” criteria collapse to the same hop count, and the tiebreaking happens to favor balance.

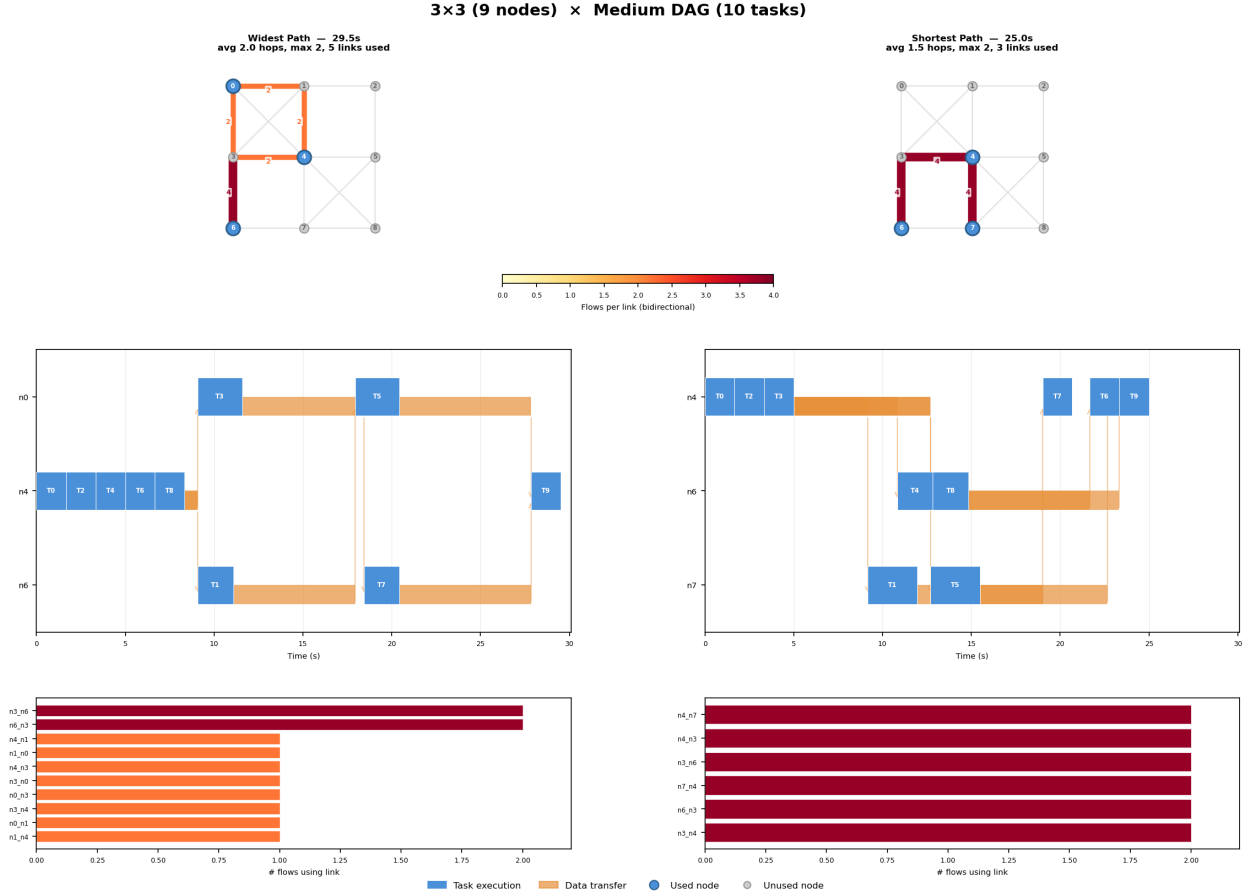


Figure 5: $3 \times 3 \times$ Medium DAG. Widest path scatters traffic through $n0$ and $n6$ via 10 different links; shortest path uses 6 links concentrated around $n4$'s neighbors.

5.3 Scaling Behavior

The widest-path penalty grows with both network size and DAG complexity:

- **Network size:** Larger grids have longer “widest” routes (up to 8 hops on 4×4), amplifying the hop-count penalty.
- **DAG complexity:** More tasks produce more concurrent transfers, all competing for the same concentrated corridor.

On the 4×4 grid, widest path consistently produces makespans $5\text{--}35\times$ longer than shortest path.

6 Conclusion

Under CSMA/Bianchi interference on wireless mesh networks, shortest-path routing significantly outperforms widest-path routing for DAG-scheduled task graphs. The key insight is that single-flow optimal routing (maximize bottleneck bandwidth) becomes counterproductive when multiple flows compete for shared wireless airtime. Shortest-path routing minimizes per-transfer link occupancy,

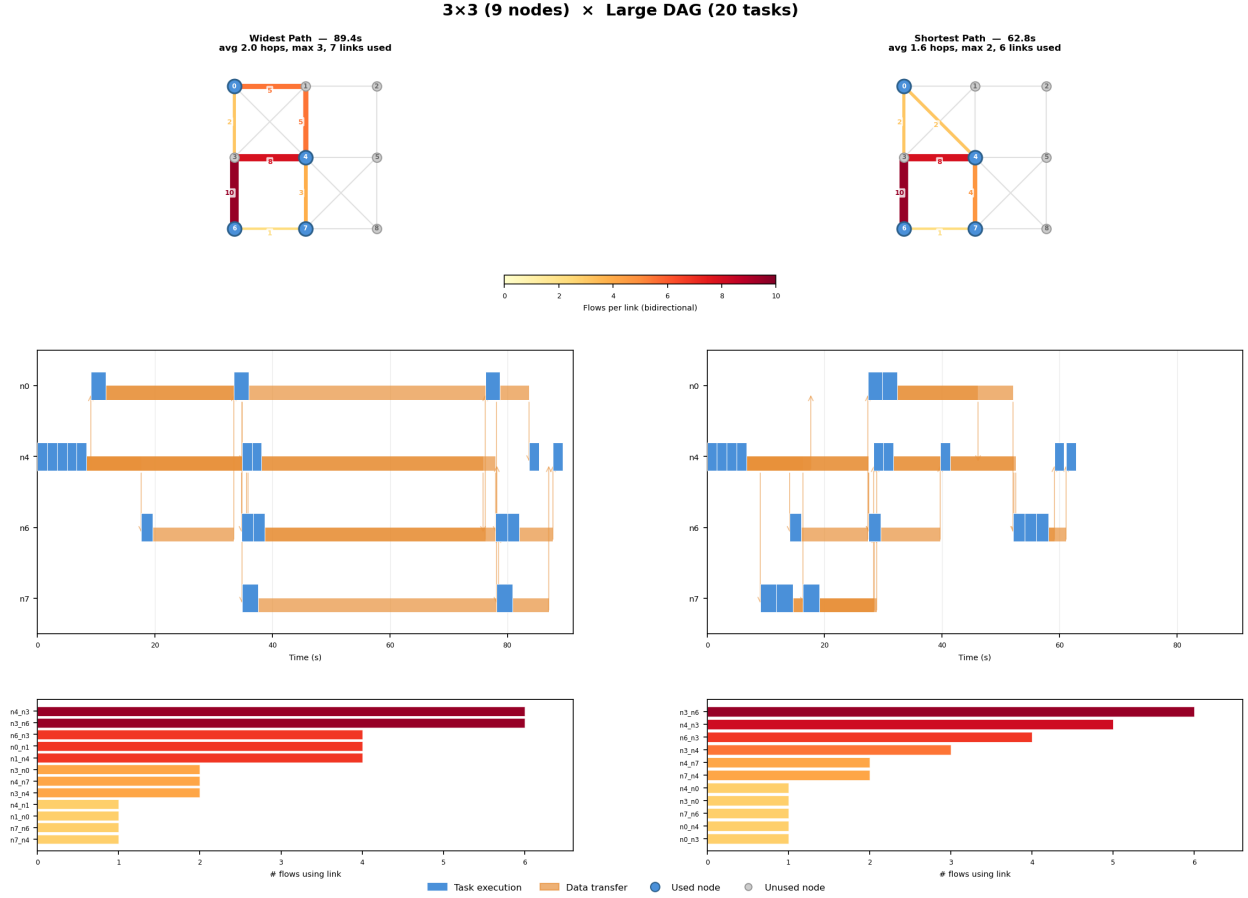


Figure 6: $3 \times 3 \times$ Large DAG. Widest path creates a congested $n4$ – $n3$ – $n6$ corridor with 6 flows per link; some transfers include 3-hop routes.

reduces the contention footprint, and distributes traffic across more diverse paths. Future work could explore contention-aware routing that jointly optimizes bandwidth and expected interference load.

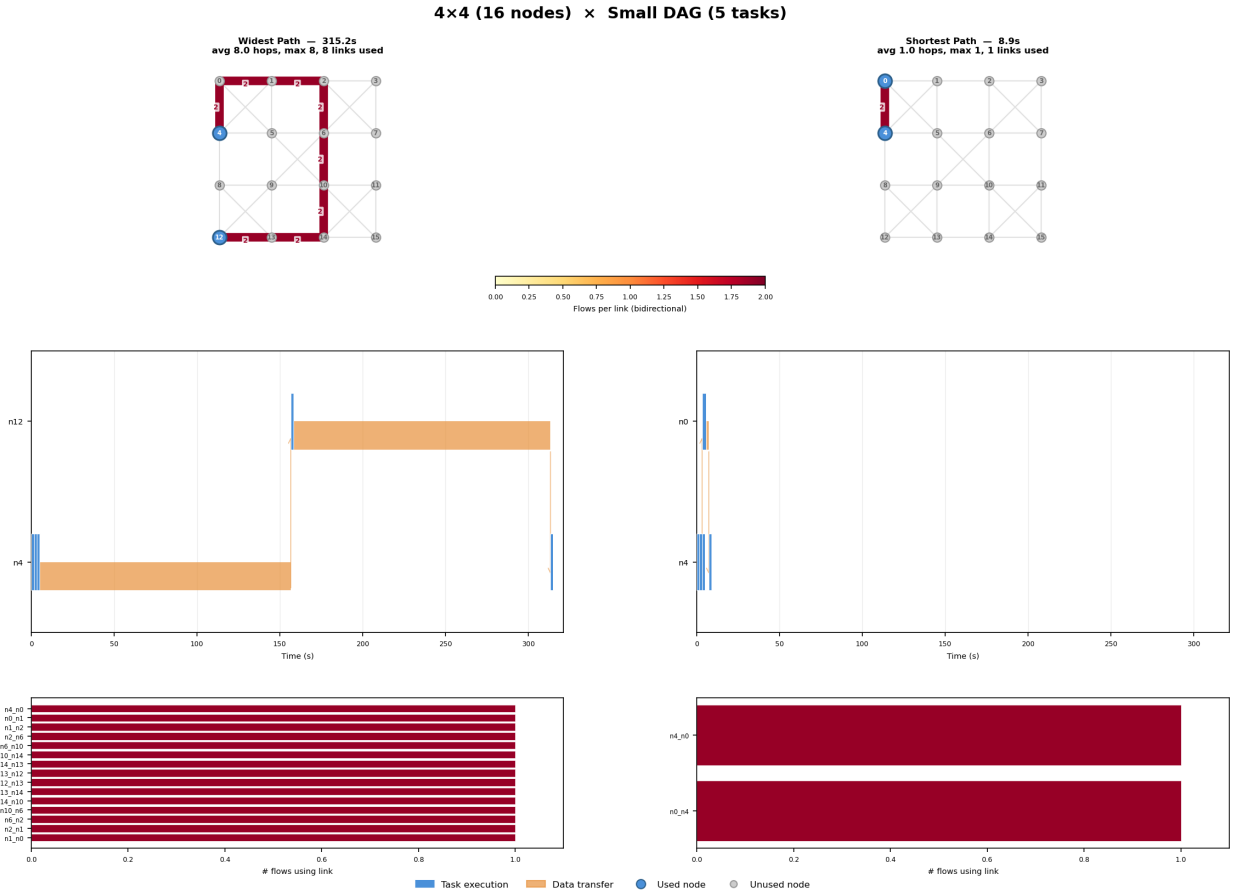


Figure 7: $4 \times 4 \times$ Small DAG. Widest path takes an 8-hop perimeter route to n12; shortest path uses a 1-hop direct link to n0. The $35\times$ slowdown is the most extreme case.

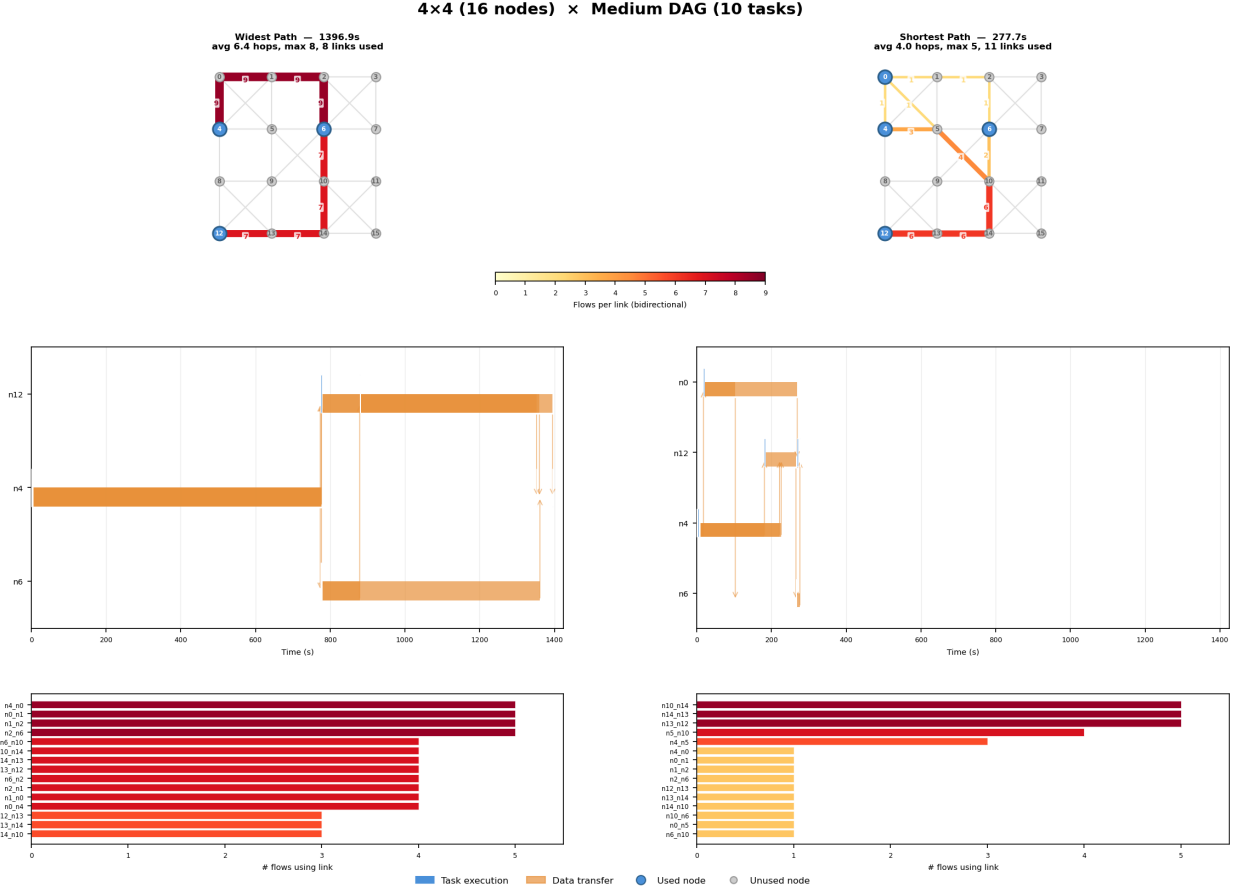


Figure 8: $4 \times 4 \times$ Medium DAG. Widest path funnels all 10 transfers through the same 8-hop perimeter corridor; shortest path splits traffic across two independent 4–5 hop paths.

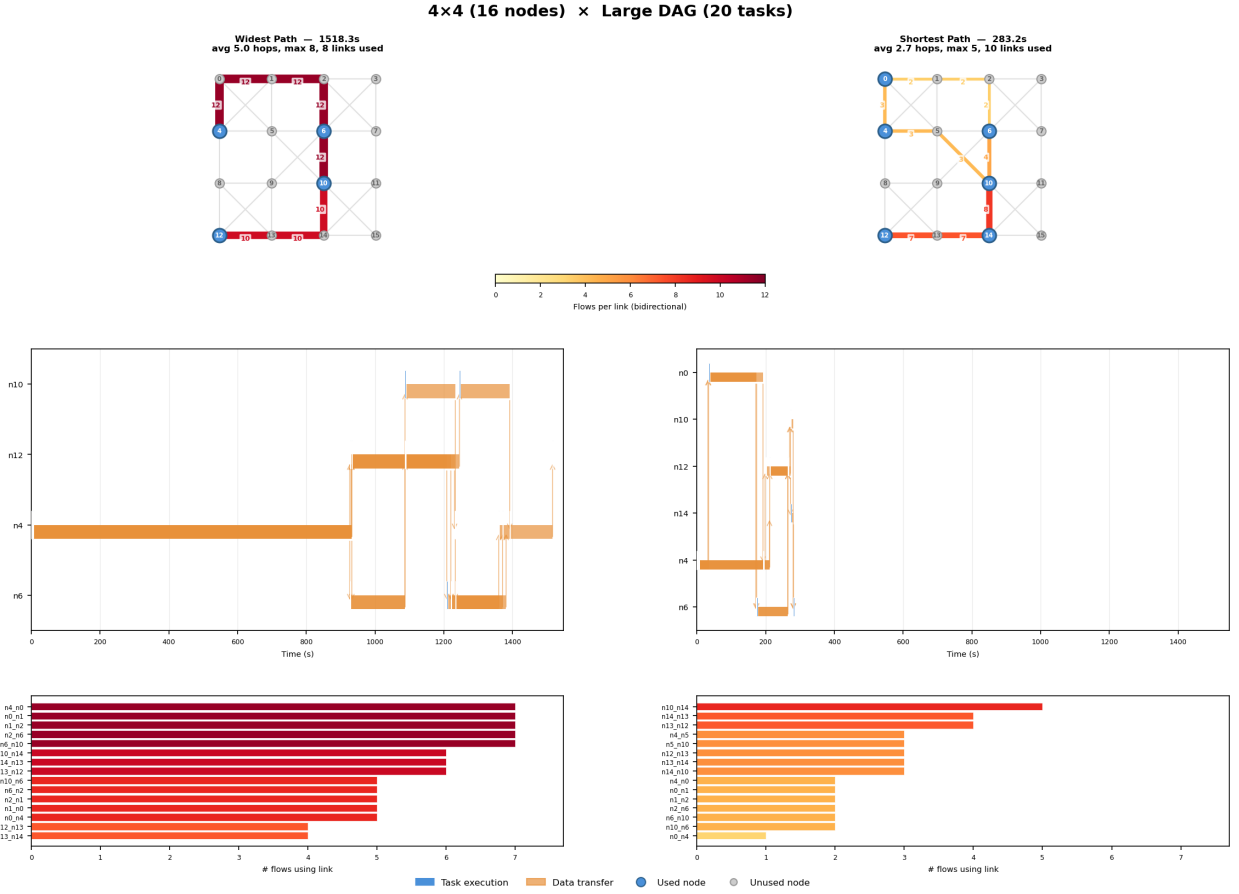


Figure 9: $4 \times 4 \times$ Large DAG. Widest path loads perimeter links with up to 12 bidirectional flows; shortest path distributes across the right side of the grid.